

CHANDNI VERMA

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Summary

Remote sensing and geospatial professional specializing in satellite image processing, DEM-based terrain analysis, SAR interferometry, and Python-driven geospatial workflow automation. Skilled in multi-source geospatial data processing, terrain analysis, and scalable remote sensing workflow development.

Technical Skills

GIS & Remote Sensing: ArcGIS, QGIS, Google Earth Engine, ENVI/SARscape, SNAP, SARPROZ, ERDAS IMAGINE

Programming: Python, Streamlit, MATLAB, R, Rasterio, GeoPandas, NumPy, SciPy, Matplotlib, TopoToolbox

Core Competencies: Optical & SAR Image Processing, DInSAR & PS-InSAR, DEM & Terrain Analysis, Watershed Delineation, Geospatial Automation, Change Detection, Raster & Vector Processing

Achievements & Scholarship

- GATE 2025 AIR 153 (Geomatics)
- GATE 2024 AIR 239 (Geomatics)
- GATE 2022 AIR 763 (Geology)
- Recipient AICTE M.Tech GATE Scholarship (2022–2024)
- Research presentations at IIRS Academia Meet (IAM) & Chandigarh Science Congress (CHASCON)

Languages

Hindi, English, Punjabi

Professional Experience

CSIR-Central Scientific Instruments Organization (CSIO)
Junior Research Fellow

Sep 2024 – Mar 2026 | Chandigarh, India

A comprehensive Geomorphic Anomaly Investigation over Sutlej Basin India

- Developed GeoProfiler, an automated Python tool for DEM-based topographic profile extraction and terrain analysis.
- Conducted geomorphic, hydrological, and morphotectonic analyses using Python, MATLAB, TopoToolbox, and GIS tools.
- Performed watershed delineation, drainage analysis, terrace mapping, longitudinal profile extraction, and χ -index assessment.
- Conducted field validation and contributed to technical reporting and documentation.

Key Projects

GeoProfiler | Python, Streamlit, Rasterio, GeoPandas

- Developed and deployed an open-source web application for extracting and visualizing topographic line and swath profiles from DEMs.
- Implemented DEM visualization, hillshade rendering, CRS handling, GeoJSON/Shapefile support, and PNG, PDF, and CSV exports.
- Designed an intuitive browser-based workflow that streamlines terrain analysis and reduces repetitive GIS processing for geomorphological and topographic investigations.

GitHub: github.com/chandnivermageo/GeoProfiler-WebApp 
Web App: geoprofiler.streamlit.app 

Groundwater Depletion & Surface Deformation Analysis | GRACE, Sentinel-1, Aquifer System Compaction

- Analyzed GRACE groundwater-storage anomalies and Sentinel-1 deformation data to investigate groundwater depletion and land subsidence.
- Processed 151 Sentinel-1 SLC scenes using DInSAR and PS-InSAR workflows and performed aquifer compaction modeling for deformation assessment.

Reservoir Sedimentation & Storage Capacity Assessment | Sentinel-1, GEE

- Assessed reservoir sedimentation using Sentinel-1 SAR GRD imagery and multi-temporal water-spread analysis in Google Earth Engine.
- Integrated SAR-derived water extent with India WRIS reservoir-level data to estimate storage-capacity changes and sedimentation impacts.

Snow-Cover Delineation & Change Detection | Python, Landsat 8, DEM

- Developed an automated Python workflow for snow-cover delineation and temporal change analysis using Landsat 8 imagery and DEM integration.
- Conducted raster analysis, change detection, and elevation-based snow distribution assessment.

GitHub: github.com/chandnivermageo/Snow-Cover-Change-Detection 

Leadership & Activities

- Winner (Mixed Table Tennis & Badminton), CSIR–CSIO Sports Tournament, 2025
- Conference Operations Team Member, International Conference on Land Surface Deformation, CSIR–CSIO (Dec 2024)

Interests

- Singing, Guitar, Badminton & Table Tennis

Education

M.Tech in Remote Sensing and GIS (Geosciences)

Indian Institute of Remote Sensing (IIRS), ISRO
Aug 2022 – July 2024 | Dehradun, India

M.Sc in Geology

Panjab University
July 2014 – June 2016 | Chandigarh, India

B.Sc in Geology

Panjab University
July 2011 – June 2014 | Chandigarh, India